

# Robust Breast Ultrasound Classification Through Dataset Auditing, Preprocessing Standardization, and Ensemble Fusion

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## Abstract

Deep learning models for breast ultrasound classification frequently report competitive benchmark performance on individually collected private or public datasets, yet this paper demonstrates that such results remain limited by dataset artifacts, inconsistent preprocessing, and weak external generalization. This study investigates breast ultrasound malignancy classification using an artifact-aware pipeline that combines dataset auditing, controlled preprocessing, backbone comparison, and ensemble fusion. Eight public breast ultrasound datasets were audited for duplicate or near-duplicate images, color Doppler overlays, and measurement markers. The audit revealed substantial non-anatomical content across datasets, including extensive duplicates in BUSDB2023, universal Doppler overlays in BUS-UCLM, and universal measurement markers in BRA, indicating a high risk of shortcut learning and biased evaluation. Three convolutional backbones, VGG-16, ResNet-50, and EfficientNetV2S, were evaluated under raw, median-filtered, CLAHE-enhanced, and full artifact-reduction preprocessing conditions using a unified training and threshold-selection protocol. Results showed that preprocessing strongly affected sensitivity, specificity, and AUC, often as much as architecture selection. Models that achieved high single-dataset performance showed reduced AUC under multi-dataset evaluation, highlighting limited generalizability. Ensemble experiments demonstrated that ResNet-50 and EfficientNetV2S provide complementary decision behavior: weighted fusion achieved the highest specificity, while attention-based fusion achieved the highest sensitivity and F1-score. These findings suggest that deployable breast ultrasound AI should be developed as a complete clinical pipeline rather than as an isolated model-selection task. Dataset curation, artifact removal, transparent preprocessing, calibration, and clinically aligned threshold selection are essential for reliable evaluation and future multi-center deployment.

**Keywords:** breast ultrasound; breast cancer classification; dataset auditing; image artifacts; preprocessing; ensemble learning; sensitivity; specificity.

## 1 Introduction

Breast cancer remains one of the most common malignancies among women worldwide, and early detection is essential for improving clinical outcomes [World Health Organization, 2023]. Ultrasound is an important modality in breast assessment because it is non-invasive, real-time, comparatively inexpensive, and useful for evaluating dense breast tissue [Al-Dhabyani et al., 2020]. These advantages have motivated extensive research on computer-aided breast lesion analysis using convolutional neural networks (CNNs), including VGG-style models, residual networks, and compound-scaled architectures [Simonyan and Zisserman, 2015, He et al., 2016, Lee and Choi, 2025]. However, a critical gap exists between reported benchmark performance and practical clinical deployability. Many studies report high accuracy or area under the receiver-operating-characteristic curve (AUC) on public datasets, but these results are often obtained under favorable experimental conditions, such as single-dataset evaluation, limited external validation, and

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inconsistent preprocessing [Zhang et al., 2020a, Thomas et al., 2023]. In clinical practice, ultrasound images vary across scanners, operators, hospitals, annotation conventions, and storage pipelines. A model that performs well on a curated benchmark may therefore fail when exposed to images acquired under different conditions. Two methodological issues motivate this study. First, public breast ultrasound datasets can contain non-anatomical visual information, including measurement calipers, color Doppler overlays, text labels, annotation marks, and duplicate or near-duplicate images. If these cues are correlated with dataset source or diagnostic label, a classifier may learn shortcut features that inflate retrospective performance without improving lesion characterization. Second, architecture comparisons are frequently confounded by uncontrolled preprocessing and dataset composition. Differences attributed to model design may instead reflect artifact prevalence, class imbalance, contrast normalization, image resolution, or annotation practice. The clinical consequence is most evident in the sensitivity–specificity trade-off. A classifier with high specificity but poor sensitivity may appear acceptable according to aggregate metrics, yet it would be unsafe for screening if it misses many malignant lesions. Conversely, very high sensitivity with poor specificity can overload radiologists with false positives and unnecessary follow-up. Therefore, reliable evaluation requires reporting threshold-dependent clinical metrics under clearly specified preprocessing conditions, not only headline AUC. This paper investigates breast ultrasound classification through a pipeline that combines dataset auditing, artifact-aware preprocessing, controlled backbone evaluation, and ensemble fusion. The principal contributions are as follows:

1. **Dataset audit.** Eight public breast ultrasound datasets are quantitatively inspected for duplicate or near-duplicate images, color Doppler overlays, markers with color indicating cancer, and measurement markers that may bias training and evaluation.
2. **Preprocessing analysis.** Raw, median-filtered, CLAHE-enhanced, and full artifact-reduction pipelines are compared under a common training and threshold-selection protocol.
3. **Single-Dataset vs. Multi-Dataset Comparison.** We show that models achieving 88-95% AUC on single datasets drop to 70-83% AUC when tested on multi-dataset combinations, revealing substantial overfitting and dataset-specific bias.
4. **Complementary ensemble learning.** Averaging, weighted, stacked, and attention-based fusion strategies are evaluated to combine the high-sensitivity behavior of ResNet-50 with the stronger specificity of EfficientNetV2S.

The remainder of the paper is organized as follows. Section 2 reviews related work. Section 3 describes the datasets, preprocessing pipeline, model architectures, training protocol, and ensemble methods. Section 4 presents quantitative results. Section 5 discusses clinical implications and limitations and concludes the study.

## 2 Related Work

In this study, we considered publicly available open-source datasets suitable for further research. The literature includes eight datasets containing breast ultrasound images: BrEaST/TCIA [Pawłowska et al., 2024], BUSI2020 [Al-Dhabyani et al., 2020], BUSDB2023 [Abbasian Ardakani et al., 2023], UDIAT2021 [Yap et al., 2018], OASBUD2017 [Piotrkowska-Wróblewska et al., 2017], BUS-UCLM [Vallez et al., 2025], Y.Guo [Guo et al., 2021], and BRA [Gómez-Flores et al., 2019]. These datasets were collected from different hospitals using different imaging tools. One of the main issues with such datasets is that some images have been marked by clinicians with red color overlays (e.g., color Doppler overlays) or cross-lines used to measure the extent of a lesion. We refer to such images as *potentially marked* images, and images without such markers or annotations as *unmarked* images. One baseline framework in the literature, presented by Zhang et al. [Zhang et al., 2020b], evaluated several models on a private breast ultrasound dataset, where InceptionV3 achieved the highest AUC of 0.905, outperforming VGG16, ResNet50, and VGG19, which obtained AUC values of

0.866, 0.851, and 0.847, respectively. Notably, their reported performance relied on images containing intact medical annotations and unstandardized contrast levels, which masks the true generalizability of the models, as such images would be classified as ‘potentially marked’ rather than ‘unmarked’ under our definition. A recent study by Koshy et al. [Koshy et al., 2026] investigates and compares EfficientNetB7, DenseNet121, ConvNeXtTiny, XGBoost Ensemble, and Soft Voting Ensemble. In our replication of this methodology, we substitute EfficientNet-V2-S for the original EfficientNet-B7 backbone, to keep this baseline comparison consistent with the EfficientNet variant used throughout the remainder of this paper’s own architecture and ensemble experiments. Across all three datasets, the Soft Voting Ensemble achieved the best overall performance, suggesting that ensemble-based approaches, particularly soft voting, can improve classification performance compared with individual architectures. For example, on the BUSI dataset, it obtained an accuracy of 88.46%, precision of 88.49%, recall of 88.46%, F1-score of 88.45%, and AUC of 95.38%. Similarly, for the BUS-UCLM and UDIAT datasets, the Soft Voting Ensemble achieved accuracy, precision, recall, F1-score, and AUC values of at least 90%. These results were obtained on ‘potentially marked’ datasets, without removing markers or crosses indicating cancer regions. To assess how much results differ when applying the methodology of Koshy et al. [Koshy et al., 2026] to the eight datasets collected for this study, we evaluated the architectures under the ‘potentially marked’ scenario rather than the fully ‘marked’ one, with results shown in Tables 1 and 2, respectively. We observed that performance drops significantly when evaluated on datasets lacking medical annotation attributes, indicating a high risk of model bias and data leakage, as networks may learn to classify based on artifacts rather than actual pathology. Datasets with available patient identifiers were split into training and test sets at the patient level to avoid leakage; datasets lacking this metadata were assigned entirely to the test set (stratified by cancer/non-cancer class), and all eight datasets were then combined into a single pooled cohort. Table 1 reports all five architectures evaluated under the original,

**Table 1:** Performance of the five baseline classification models, implemented following the methodology of Koshy et al. [Koshy et al., 2026], on the original datasets before artifact removal.

| Model                | Accuracy | F1-Score | Sensitivity | Specificity | AUC    |
|----------------------|----------|----------|-------------|-------------|--------|
| Soft_Voting_Ensemble | 0.8924   | 0.8522   | 0.8305      | 0.9293      | 0.9680 |
| ConvNeXt_Tiny        | 0.9051   | 0.8718   | 0.8644      | 0.9293      | 0.9632 |
| DenseNet121          | 0.8671   | 0.8073   | 0.7458      | 0.9394      | 0.9539 |
| EfficientNetV2S      | 0.8734   | 0.8276   | 0.8136      | 0.9091      | 0.9462 |
| XGBoost_Ensemble     | 0.5696   | 0.1282   | 0.0847      | 0.8586      | 0.4181 |

pre-artifact-removal setting. After applying the artifact-removal and preprocessing pipeline described in Section 3, EfficientNetV2S and the Soft Voting Ensemble retained the strongest performance, while the remaining three architectures showed substantially greater degradation. These two models were therefore carried forward to the subsequent ensemble experiments (Table 2); the XGBoost Ensemble was additionally unstable even before artifact removal (AUC 0.4181), indicating it failed to learn a usable decision boundary on this data. After artifact removal, both models show a marked drop in performance across all metrics

**Table 2:** Performance of the two best-performing models after artifact removal and preprocessing.

| Model               | Accuracy | F1-Score | Sensitivity | Specificity | AUC    |
|---------------------|----------|----------|-------------|-------------|--------|
| EfficientNetV2S     | 0.7421   | 0.6734   | 0.6837      | 0.7792      | 0.7662 |
| SoftVoting_Ensemble | 0.7262   | 0.6230   | 0.5816      | 0.8182      | 0.7435 |

relative to Table 1, indicating that the original scores were inflated by non-anatomical artifacts rather than true lesion signal.

### 3 Materials and Methods

#### 3.1 Dataset Audit

The eight aforementioned public breast ultrasound datasets were audited to verify dataset composition and quantify visible sources of potential bias. For each dataset, images were re-examined to estimate the total number of images, the proportion of cancer cases, duplicate or near-duplicate image groups, color Doppler overlays, and measurement markers. The audit was designed as a quality-control step before model training, with the goal of identifying possible leakage and shortcut-learning signals.

**Duplicate detection.** Perceptual hashing was used as an initial screen for near-duplicate content. Each image was resized to  $8 \times 8$  pixels, converted to grayscale, and represented by a binary hash

$$h(i) = \begin{cases} 1, & I(i) > \mu, \\ 0, & \text{otherwise,} \end{cases} \quad (1)$$

where  $I(i)$  is the intensity of pixel  $i$  in the resized image and  $\mu$  is the mean intensity. Candidate pairs with Hamming distance below 5 were then validated using the structural similarity index measure (SSIM). Pairs with SSIM greater than 0.95 were treated as duplicate or near-duplicate images.

**Artifact detection.** Color overlays, including Color Doppler visualizations and colored annotations (e.g., red, blue, or yellow boxes and markers), were detected by analyzing differences between the RGB color channels. Since grayscale ultrasound images exhibit similar intensities across all three channels, images with large channel differences were flagged as containing color overlays.

#### 3.2 Preprocessing Pipeline

The preprocessing pipeline was designed to reduce non-anatomical information while preserving lesion morphology. Colored overlays and markers were first segmented in HSV color space. Red regions were defined by  $H \in [0, 10] \cup [160, 180]$  with saturation and value greater than 100, and blue regions were defined by  $H \in [100, 130]$ . The resulting binary masks were refined with a  $5 \times 5$  morphological kernel and inpainted using Telea’s fast marching method [Telea, 2004]. Median filtering with a  $5 \times 5$  kernel was then applied to reduce isolated residual artifacts while preserving boundaries. Finally, CLAHE was applied to the luminance channel in LAB color space with clip limit 2.0 and tile grid size  $8 \times 8$ . Four preprocessing conditions were evaluated: raw images, median-filtered images, CLAHE-enhanced images, and the full artifact-reduction pipeline. This design allowed the effect of each preprocessing regime to be compared under a common model and evaluation protocol. These values follow common defaults in the ultrasound and general image-restoration literature (e.g., typical CLAHE clip-limit ranges and Telea’s original parameterization) and were checked on a held-out validation subset to confirm they did not distort lesion boundaries before being fixed for all experiments.

#### 3.3 Classification Architectures

Three CNN backbones were evaluated: VGG-16, ResNet-50, and EfficientNetV2S. The backbones were paired with a common attention-augmented classification head to reduce differences introduced by the final classifier. Attention was implemented using a convolutional block attention module (CBAM) [Woo et al., 2018], which combines channel and spatial attention. For a feature tensor  $F$ , channel attention was computed using global average pooling followed by a two-layer multilayer perceptron:

$$s = \sigma(W_2 \text{ReLU}(W_1 \text{GAP}(F))), \quad \tilde{F} = s \odot F, \quad (2)$$

where  $\sigma$  denotes the sigmoid function and  $\odot$  denotes element-wise multiplication. Spatial attention was then computed using a  $7 \times 7$  convolution over concatenated average-pooled and max-pooled feature maps. The resulting representation was passed through batch-normalized fully connected layers of sizes 512 and 256 with dropout probability  $p = 0.3$ , followed by a sigmoid output neuron.

### 3.4 Training and Evaluation Protocol

All models were trained using the Adam optimizer with transfer learning from ImageNet-pretrained weights. The backbone was initially frozen while the classification head was trained with learning rate  $10^{-4}$ ; subsequent fine-tuning used learning rate  $10^{-5}$ . Training augmentation included horizontal flipping, rotations up to  $\pm 40^\circ$ , and color jitter. Class imbalance was addressed using oversampling within the training set only. Because missing malignant cases is clinically more costly than false-positive referral, training used a sensitivity-oriented loss combining focal loss [Lin et al., 2017] with a false-negative penalty:

$$\mathcal{L} = -\alpha_t(1 - p_t)^\gamma \log(p_t) + \lambda_{\text{FN}} \mathbb{E}[y(1 - \hat{p})], \quad (3)$$

where  $p_t$  is the probability assigned to the true class,  $\hat{p}$  is the predicted malignancy probability,  $y \in \{0, 1\}$  is the ground-truth label,  $\alpha_t = 0.25$ ,  $\gamma = 2.0$ , and  $\lambda_{\text{FN}} = 6.0$ . Decision thresholds were selected using Youden’s index on the ROC curve. Performance was reported using AUC, sensitivity, specificity, and F1-score.

### 3.5 Ensemble Methods

ResNet-50 and EfficientNetV2S were selected for ensemble analysis because their individual results showed complementary behavior: ResNet-50 tended to improve malignant-case detection, whereas EfficientNetV2S tended to preserve specificity. Four fusion strategies were evaluated.

**Averaging.** The predicted malignancy probability was computed as

$$\hat{p}_{\text{avg}} = \frac{\hat{p}_{\text{R}} + \hat{p}_{\text{E}}}{2}, \quad (4)$$

where  $\hat{p}_{\text{R}}$  and  $\hat{p}_{\text{E}}$  denote the ResNet-50 and EfficientNetV2S predictions.

**Weighted fusion.** Learned softmax-normalized weights were used to combine probabilities:

$$\hat{p}_{\text{w}} = w_{\text{R}}\hat{p}_{\text{R}} + w_{\text{E}}\hat{p}_{\text{E}}, \quad w_{\text{R}} + w_{\text{E}} = 1. \quad (5)$$

**Stacked fusion.** A two-layer meta-classifier with architecture  $2 \rightarrow 64 \rightarrow 32 \rightarrow 1$  was trained on frozen base-model predictions.

**Attention fusion.** Feature-level fusion was implemented using multi-head attention with eight heads. ResNet-50 and EfficientNetV2S feature vectors were projected into a common embedding space, and cross-attention was used to learn task-dependent interactions before classification. This design allows the fused model to weight complementary representations differently across images rather than using a fixed probability average. More details are shown in Figure 1.

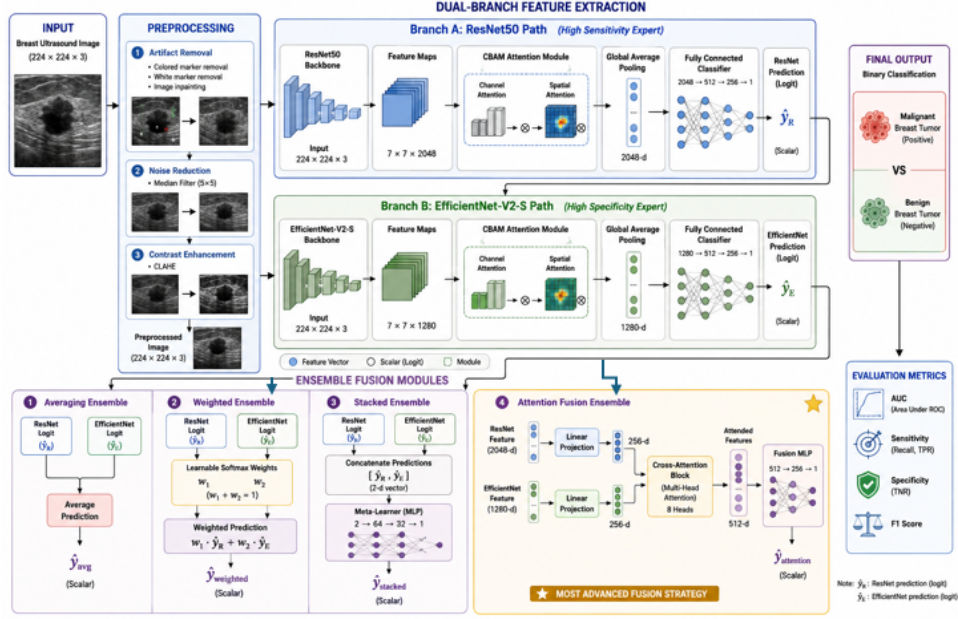
## 4 Results

### 4.1 Dataset Audit

Table 5, added at the end of this paper, summarizes the dataset audit. The results show that non-anatomical content is common across public breast ultrasound datasets. BUSDB2023 contained the largest number of



duplicate or near-duplicate groups and color Doppler overlays. BUS-UCLM contained color Doppler overlays in all audited images, and BRA contained measurement markers in all audited images. These findings indicate that image artifacts should be treated as a central part of the experimental design rather than as incidental noise. The audit has two implications for model evaluation. First, duplicate and near-duplicate images can create leakage if visually similar samples are split across training and testing. Second, overlays and markers may act as label-correlated shortcuts. Even if such cues are clinically meaningful in some contexts, their inconsistent presence across datasets makes them unreliable for a B-mode lesion-classification model intended for deployment across institutions.



**Figure 1:** Architecture of the proposed dual-branch ensemble pipeline for breast ultrasound classification. The dual-branch design, backbone selection, CBAM attention integration, and all four fusion strategies were conceived and specified entirely by the authors; a generative AI diagramming tool was used only to assist with rendering and visually refining the final figure layout. Raw ultrasound images are first processed using the three-stage preprocessing pipeline described in Section 3: HSV-based segmentation and inpainting to remove colored markers and Doppler overlays,  $5 \times 5$  median filtering to suppress residual noise, and CLAHE-based contrast enhancement to improve consistency in tissue appearance across the pooled source datasets. Each preprocessed image is then passed through two parallel convolutional backbones, ResNet-50 and EfficientNet-V2-S. Both branches incorporate a CBAM module for combined channel and spatial attention, followed by global average pooling and a dense classification head. The outputs of the two branches are subsequently combined using one of four fusion strategies evaluated in this study: simple averaging, learned weighted averaging, a trained stacking meta-classifier, or cross-attention feature fusion. The resulting fused output produces the final benign/malignant classification. Fusion performance is evaluated using AUC, sensitivity, specificity, and F1-score, as reported in Table 4.

## 4.2 Ensemble Results

Table 4 compares the ensemble strategies. All ensemble methods improved at least one clinically relevant metric relative to the individual base models, supporting the hypothesis that ResNet-50 and EfficientNetV2S provide complementary information.

### 4.2.1 Inference Latency

To assess clinical deployment feasibility, inference latency was benchmarked for each backbone and fusion strategy on a CUDA GPU, using a warm-up period followed by 100 timed forward passes per model (batch size 1,  $224 \times 224$  input). Results are reported in Table 3. As expected, ResNet50 alone is fastest, and attention fusion

**Table 3:** Inference latency per model/ensemble strategy (CUDA, batch size 1).

| Model              | Mean (ms/image) | Median (ms/image) | Throughput (img/s) |
|--------------------|-----------------|-------------------|--------------------|
| ResNet50           | 6.58            | 6.49              | 151.9              |
| EfficientNetV2S    | 17.99           | 17.61             | 55.6               |
| Averaging Ensemble | 24.97           | 24.37             | 40.0               |
| Weighted Ensemble  | 23.98           | 23.88             | 41.7               |
| Stacked Ensemble   | 24.50           | 24.50             | 40.8               |
| Attention Fusion   | 24.83           | 24.64             | 40.3               |

is among the more computationally expensive strategies due to the added cross-attention computation, though all ensemble variants remain within a similar range (24–25 ms/image) since backbone inference dominates total latency. These figures reflect GPU (CUDA) latency; CPU-only deployment timing was not evaluated in this study and is left for future work.

**Table 4:** Ensemble comparison using ResNet-50 and EfficientNetV2S. Attention fusion achieved the highest sensitivity and F1-score; weighted fusion achieved the highest specificity.

| Model                   | AUC          | Sensitivity  | Specificity  | F1           |
|-------------------------|--------------|--------------|--------------|--------------|
| ResNet-50               | 0.777        | 0.735        | 0.682        | 0.692        |
| EfficientNetV2S         | 0.810        | 0.755        | 0.695        | 0.708        |
| Averaging ensemble      | 0.811        | 0.776        | 0.714        | 0.712        |
| Weighted ensemble       | 0.822        | 0.745        | <b>0.799</b> | 0.726        |
| Stacked ensemble        | 0.813        | 0.806        | 0.701        | 0.722        |
| <b>Attention fusion</b> | <b>0.823</b> | <b>0.888</b> | 0.662        | <b>0.733</b> |

Simple averaging improved sensitivity, specificity, and F1-score relative to both base models, indicating partially independent error patterns. Weighted fusion achieved the strongest specificity, 0.799, while preserving AUC 0.822. This operating point is better suited to confirmation settings in which false positives should be reduced. Stacked fusion increased sensitivity to 0.806, but it did not match the AUC or specificity of weighted fusion. Attention fusion produced the most screening-oriented behavior, with sensitivity 0.888 and F1-score 0.733. This sensitivity represents a substantial improvement over the individual ResNet-50 and EfficientNetV2S models. The trade-off is lower specificity than weighted fusion, which may be acceptable when the system is used as a second reader or triage aid rather than as an autonomous diagnostic device. This ablation reflects preprocessing sensitivity at inference time only, using backbones trained on the full pipeline rather than retrained per condition. We attempted a fully independent ablation, retraining each backbone from scratch under all four preprocessing conditions, but this coincided with corrections to the dataset composition and an upgrade of the Python/deep-learning library environment. The resulting AUC values were inflated and inconsistent across conditions, itself illustrating this paper’s central argument: reported AUC is sensitive to the full experimental pipeline, including dataset version and software environment, not only to architecture and preprocessing choice. Since these runs were not obtained under a fixed, reproducible pipeline, we omit their numeric values to avoid overstating the ablation’s conclusions. A corrected ablation, run under a single fixed dataset version and environment, is planned as immediate follow-up work.

## 5 Discussion and Conclusion

This study demonstrates that robust breast ultrasound classification requires a holistic pipeline approach rather than isolated model selection, as systematic dataset audits revealed that public cohorts contain significant non-anatomical artifacts (e.g., duplicates, calipers, and Doppler overlays). Our experiments show that artifact-suppressing preprocessing shifts clinical operating characteristics as strongly as backbone

**Table 5:** Combined Ultrasound Public Dataset audit results. Counts summarize visible duplicate or near-duplicate groups, color Doppler overlays, and measurement markers identified during quality control.

| Metric              | BrEaST/TCIA | BUSI | BUSDB | UDIAT | OASBUD | BUS-UCLM | Y.Guo | BRA  |
|---------------------|-------------|------|-------|-------|--------|----------|-------|------|
| Total images        | 252         | 780  | 232   | 163   | 100    | 264      | 192   | 1875 |
| Cancer (%)          | 38.9        | 32.5 | 53.0  | 33.1  | 52.0   | 34.1     | 12.0  | 32.4 |
| Duplicates          | 0           | 98   | 1     | 4     | 0      | 21       | 5     | 0    |
| Doppler overlays    | 0           | 6    | 0     | 0     | 0      | 36       | 2     | 0    |
| Measurement markers | 0           | 749  | 361   | 0     | 101    | 11       | 5     | 1875 |

selection, revealing whether networks rely on true lesion morphology or shortcut signals. Furthermore, leveraging the complementary inductive biases of ResNet-50 (which enhances malignancy detection) and EfficientNetV2S (which preserves specificity) through learned fusion yields clinically tailorable workflows; specifically, cross-attention fusion maximizes sensitivity whereas weighted fusion optimizes specificity. While these findings underscore the value of pipeline-wide optimization, several limitations remain. This work relies on retrospective public datasets, employs heuristic parameters for preprocessing (such as CLAHE limits) and artifact detection, and introduces additional computational overhead via attention-based fusion. Our attempted preprocessing ablation (Section 4) further underscored this fragility in practice, as dataset and software-environment changes alone produced unstable AUC shifts comparable in magnitude to those attributed to preprocessing itself. Additionally, despite incorporating CBAM and attention mechanisms, the interpretability of these ensemble decisions requires further refinement prior to clinical adoption. Future work will validate this framework on prospective multicenter cohorts with consistent acquisition metadata, develop adaptive quality-control models, incorporate calibration-aware thresholding, extend the system from binary to multi-class, BIRADS-oriented decision support, and conduct a fully independent per-condition preprocessing ablation that re-trains each backbone separately to isolate its contribution to performance. We are also collecting a new breast ultrasound dataset with a clinical partner hospital – prospectively acquired, single-institution, with consistent imaging protocols and verified groundtruth labels – to validate this framework beyond the retrospective public datasets used here.

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